

ECON622 Project 1: Simple Linear Regression

S&P 500 Returns and US Unemployment rate

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```
knitr::opts_chunk$set(  
  echo = FALSE,  
  fig.align = "center",  
  fig.height = 5,  
  fig.width = 7,  
  message = FALSE,  
  warning = FALSE  
)
```

1. Introduction

This study examines how S&P 500 stock returns (X) affect US Monthly unemployment rate (Y) from January 1948 to August 2025.

a. Variable Choice

Y variable: Monthly unemployment rate (unemp)

X variable: S&P 500 returns (sp500)

We use unemployment as a continuous macro indicator of labor market slack. Market levels are forward-looking; using lagged S&P 500 helps test whether past market conditions are associated with next month's unemployment.

Time Period: Jan 1949 to August 2025 (75.6 years)

Why this period:

The time period from January 1949 to August 2025 (75.6 years) provides a uniquely comprehensive view of the U.S. economy and financial markets. It spans multiple economic cycles, including major recessions such as the dot-com crash (2001), the global financial crisis (2008–2009), and the COVID-19 pandemic (2020) with COVID serving as a particularly interesting case study of how sudden shocks impact both stock performance and unemployment. Covering more than three-quarters of a century, the dataset delivers a large sample size of 900+ monthly observations, which strengthens statistical reliability. Importantly, it combines both historical depth and modern market dynamics, making the findings relevant for understanding long-term relationships while still reflecting contemporary economic behavior.

Project Plan: This is project 1 with simple regression ($Y_t = \alpha + \beta X_t + \varepsilon_t$). Project 2 will add more variables.

b. Literature Review

The relationship between economic cycles and stock market returns has been extensively studied in both theoretical and empirical literature.

Theoretical Foundation:

Financial market theory suggests that stock prices incorporate forward-looking information about economic conditions. Stock market declines often precede economic recessions as investors anticipate deteriorating corporate earnings, reduced consumer spending, and tightening credit conditions. The wealth effect and confidence channels provide theoretical mechanisms through which stock market performance influences broader economic activity and Monthly unemployment rate.

Empirical Evidence:

Green A, et al. (2024)^[3], in their study “Identifying Economic Factors Affecting Unemployment Rates”, examined nearly two decades of U.S. data to understand how macroeconomic indicators influence labor market conditions. Using regression analysis, they found that the S&P 500 index is negatively correlated with the unemployment rate, meaning that periods of strong stock market performance are typically accompanied by lower unemployment.

This relationship highlights the role of the stock market as a forward-looking indicator of economic health. Rising S&P 500 returns often reflect improved corporate profitability and investment, which translates into greater hiring and job creation. Conversely, a declining stock market signals weaker business conditions, reduced investment, and ultimately higher unemployment rates.

COVID-19 Period as a Case Study: The COVID-19 pandemic period (2020-2021) represents a particularly interesting case within our sample. Unlike typical recessions driven by financial or economic imbalances, the COVID-19 recession was caused by an external public health crisis that led to unprecedented government-mandated economic shutdowns.

2. Data

```
library(tidyverse)
library(readr)
library(knitr)
library(ggplot2)
library(zoo)
library(lubridate)
library(kableExtra)
```

a. Data Sources

Files: SP500.csv and UNRATE.csv

Period: January 1948 to August 2025

Frequency: Monthly

Variables: sp500: S&P 500 returns (X variable) unemp: Monthly unemployment rate (Y variable)

```
# Load data
sp500_raw <- read_csv("SP500.csv")
unemp_raw <- read_csv("UNRATE.csv")

# --- Original cleaning ---
names(sp500_raw)[1] <- "date"
names(sp500_raw)[2] <- "sp500"
names(unemp_raw)[1] <- "date"
names(unemp_raw)[2] <- "unemp"

sp500_raw$date <- as.Date(lubridate::parse_date_time(
  sp500_raw$date,
  orders = c("mdy", "ymd", "dmy", "Y-m", "m-Y", "b-Y", "Y-b", "m/Y", "Y/m", "bY", "Yb")
))
```

```

unemp_raw$date <- as.Date(lubridate::parse_date_time(
  unemp_raw$date,
  orders = c("ymd", "dmy", "Y-m", "m-Y", "b-Y", "Y-b", "m/Y", "Y/m", "bY", "Yb")
))

start_date <- as.Date("1948-01-01")
max_date <- as.Date("2025-08-01")
sp500_raw <- sp500_raw[sp500_raw$date >= start_date, ]
unemp_raw <- unemp_raw[unemp_raw$date >= start_date, ]

sp500_raw$ym <- zoo::as.yearmon(sp500_raw$date)
unemp_raw$ym <- zoo::as.yearmon(unemp_raw$date)

sp500_m <- sp500_raw |>
  dplyr::arrange(date) |>
  dplyr::group_by(ym) |>
  dplyr::slice_tail(n = 1) |>
  dplyr::ungroup() |>
  dplyr::select(ym, sp500)

unemp_m <- unemp_raw |>
  dplyr::arrange(date) |>
  dplyr::group_by(ym) |>
  dplyr::slice_tail(n = 1) |>
  dplyr::ungroup() |>
  dplyr::select(ym, unemp)

# --- Build combined dataset for modeling ---
data <- dplyr::inner_join(sp500_m, unemp_m, by = "ym") |>
  dplyr::mutate(date = as.Date(ym)) |>
  dplyr::filter(date <= max_date)

to_num <- function(x) {
  x <- as.character(x)
  x <- gsub(",", "", x)
  suppressWarnings(as.numeric(x))
}
data$sp500 <- to_num(data$sp500) # S\&P 500 level
data$unemp <- to_num(data$unemp) # Unemployment rate

# --- Predictive variables ---
data <- data |>
  dplyr::arrange(date) |>
  dplyr::mutate(
    # Next year's changes (dependent vars)
    unemp_FG = 100 * (log(dplyr::lead(unemp, 12)) - log(unemp)), # unemployment forward growth
    sp500_FG = (dplyr::lead(sp500, 12) - sp500) / sp500 * 100, # forward 12m return (%)

    # Past year's changes (predictors)
    unemp_G = 100 * (log(unemp) - log(dplyr::lag(unemp, 12))),
    sp500_G = (sp500 - dplyr::lag(sp500, 12)) / dplyr::lag(sp500, 12) * 100,

    # COVID dummy

```

```

    covid = as.integer(date >= as.Date("2020-03-01") & date <= as.Date("2021-06-30"))
  )

# Drop NAs
data <- stats::na.omit(data)

# Quick check
cat("Rows after processing:", nrow(data),
    "| First:", format(min(data$date)),
    "| Last:", format(max(data$date)), "\n")

```

```
## Rows after processing: 908 | First: 1949-01-01 | Last: 2024-08-01
```

b. Data Descriptions

In this project, we analyze a linear regression model with Unemployment Rate (dependent variable) and S&P 500 Returns (regressor). All data used in the analysis are obtained from public sources.

The details are given below:

unemp (Y): The civilian unemployment rate measures the percentage of the labor force that is unemployed but actively seeking work. High unemployment is often associated with recessions, making it a useful proxy for economic downturns. The data are obtained from FRED^[1]. Frequency: monthly. Units: Percent (%).

sp500 (X): Monthly values of the S&P 500 index, sourced from Yahoo Finance^[2]. The sample period captures major economic cycles. Units are index points, presented as a time series.

Lag Structure and Rationale:

A 12-month lag is applied to the S&P 500 returns and unemployment variables to build a forward-looking predictive model. This approach allows us to analyze how past stock market and labor market changes influence future unemployment outcomes one year ahead, reflecting the notion that financial markets often anticipate economic conditions before they appear in employment data.

During processing, observations with missing values resulting from the lagging and leading calculations were removed. Specifically:

The first 12 months of the dataset were dropped because lagged values of S&P 500 and unemployment could not be computed.

The last 12 months were dropped because forward-looking growth variables (unemp_FG and sp500_FG) could not be calculated.

This ensures that all remaining observations have complete data for both predictors and dependent variables, resulting in a clean and consistent dataset suitable for regression analysis.

Sample Information

Sample size: 908 observations

Period: Jan 1949 to Aug 2024

COVID observations: 16 (1.8% of sample)

Descriptive Statistics

```
# Select relevant variables
DATA <- data[, c("unemp", "sp500")] # Y first, X second

# Summary statistics table
stats_table <- data.frame(
  Variable = c("Monthly Unemployment Rate (Y)", "S&P 500 Returns (X)"),
  Min = round(c(min(DATA$unemp), min(DATA$sp500)), 4),
  "1st Qu." = round(c(quantile(DATA$unemp, 0.25), quantile(DATA$sp500, 0.25)), 4),
  Median = round(c(median(DATA$unemp), median(DATA$sp500)), 4),
  Mean = round(c(mean(DATA$unemp), mean(DATA$sp500)), 4),
  "3rd Qu." = round(c(quantile(DATA$unemp, 0.75), quantile(DATA$sp500, 0.75)), 4),
  Max = round(c(max(DATA$unemp), max(DATA$sp500)), 4),
  SD = round(c(sd(DATA$unemp), sd(DATA$sp500)), 4)
)

# Fix column names to remove X prefix
colnames(stats_table) <- c("Variable", "Min", "1st Qu.", "Median", "Mean", "3rd Qu.", "Max", "SD")

# Correlation
cor_table <- round(cor(DATA), 4)
rownames(cor_table) <- c("Monthly Unemployment Rate (Y)", "S&P 500 Returns (X)")
colnames(cor_table) <- c("Monthly Unemployment Rate (Y)", "S&P 500 Returns (X)")

# Display statistics table with styling
kable(stats_table,
  caption = "Summary Statistics for Monthly Unemployment Rate and S\\&P 500 Returns \\label{tab:des",
  booktabs = TRUE,
  escape = FALSE,
  align = "lcccc") %>%
  kable_styling(latex_options = c("HOLD_position", "striped"),
    font_size = 11,
    full_width = FALSE,
    stripe_color = "blue!5!white",
    position = "center") %>%
  row_spec(0, bold = TRUE, background = "blue!50!black", color = "white")
```

Table 1: Summary Statistics for Monthly Unemployment Rate and S&P 500 Returns

Variable	Min	1st Qu.	Median	Mean	3rd Qu.	Max	SD
Monthly Unemployment Rate (Y)	2.50	4.4000	5.500	5.7159	6.725	14.8	1.706
S&P 500 Returns (X)	14.16	87.1925	247.215	814.8698	1228.915	5648.4	1122.962

```
# Display correlation table with styling
kable(cor_table,
  caption = "Correlation Matrix between Monthly Unemployment Rate and S\\&P 500 Returns \\label{tab",
  booktabs = TRUE,
  escape = FALSE,
  align = "cc") %>%
  kable_styling(latex_options = c("HOLD_position", "striped"),
```

```

font_size = 11,
full_width = FALSE,
stripe_color = "blue!5!white",
position = "center") %>%
row_spec(0, bold = TRUE, background = "blue!50!black", color = "white")

```

Table 2: Correlation Matrix between Monthly Unemployment Rate and S&P 500 Returns

	Monthly Unemployment Rate (Y)	S&P 500 Returns (X)
Monthly Unemployment Rate (Y)	1.0000	-0.1672
S&P 500 Returns (X)	-0.1672	1.0000

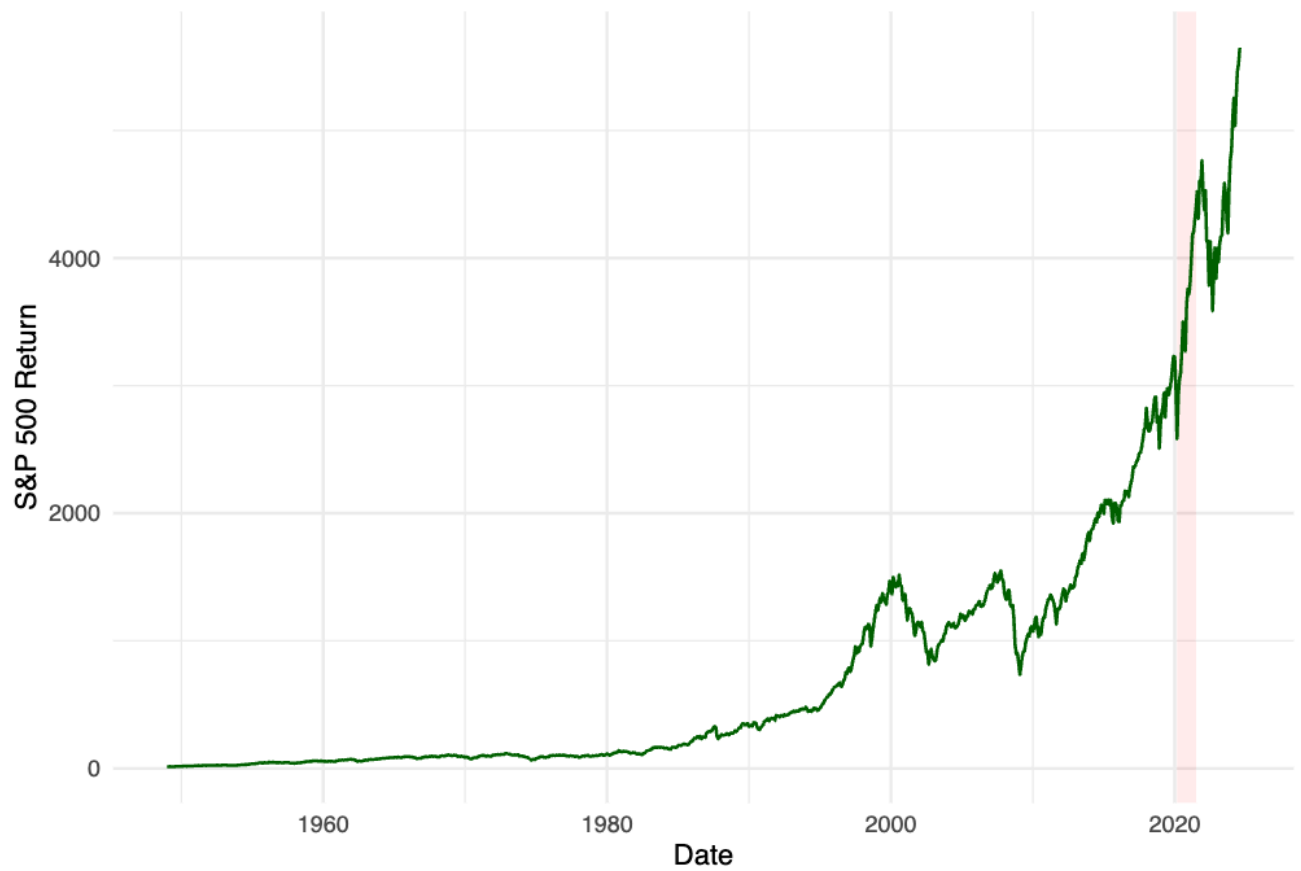
c. Plots

```

# S\&P 500 with COVID highlighting - matching your title
ggplot(data, aes(x = date, y = sp500)) +
  geom_rect(
    data = data.frame(xmin = as.Date("2020-03-01"), xmax = as.Date("2021-06-30")),
    aes(xmin = xmin, xmax = xmax, ymin = -Inf, ymax = Inf),
    inherit.aes = FALSE, fill = "red", alpha = 0.08
  ) +
  geom_line(color = "darkgreen", linewidth = 0.6) +
  labs(title = "S&P 500 Return Over Time", x = "Date", y = "S&P 500 Return") +
  theme_minimal()

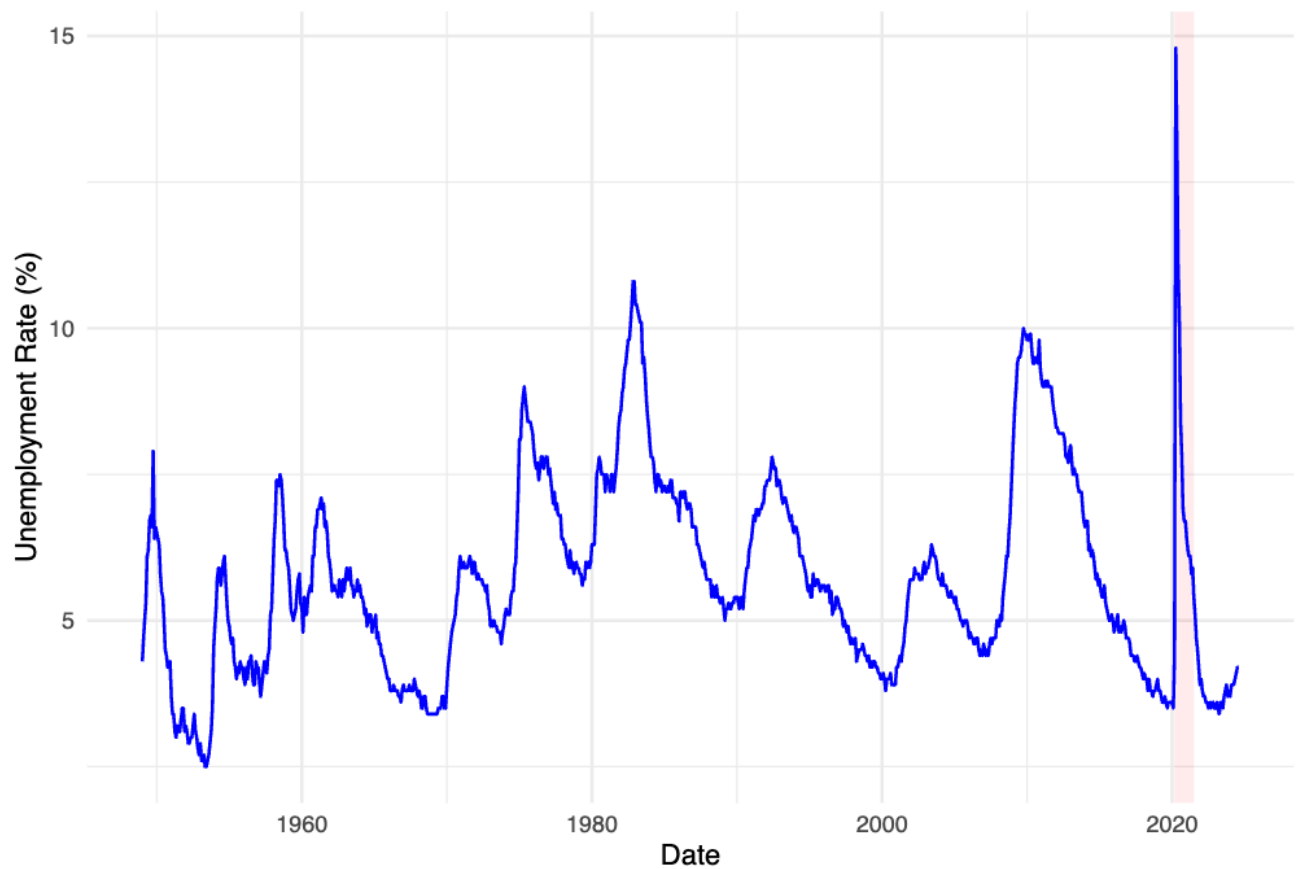
```


S&P 500 Return Over Time

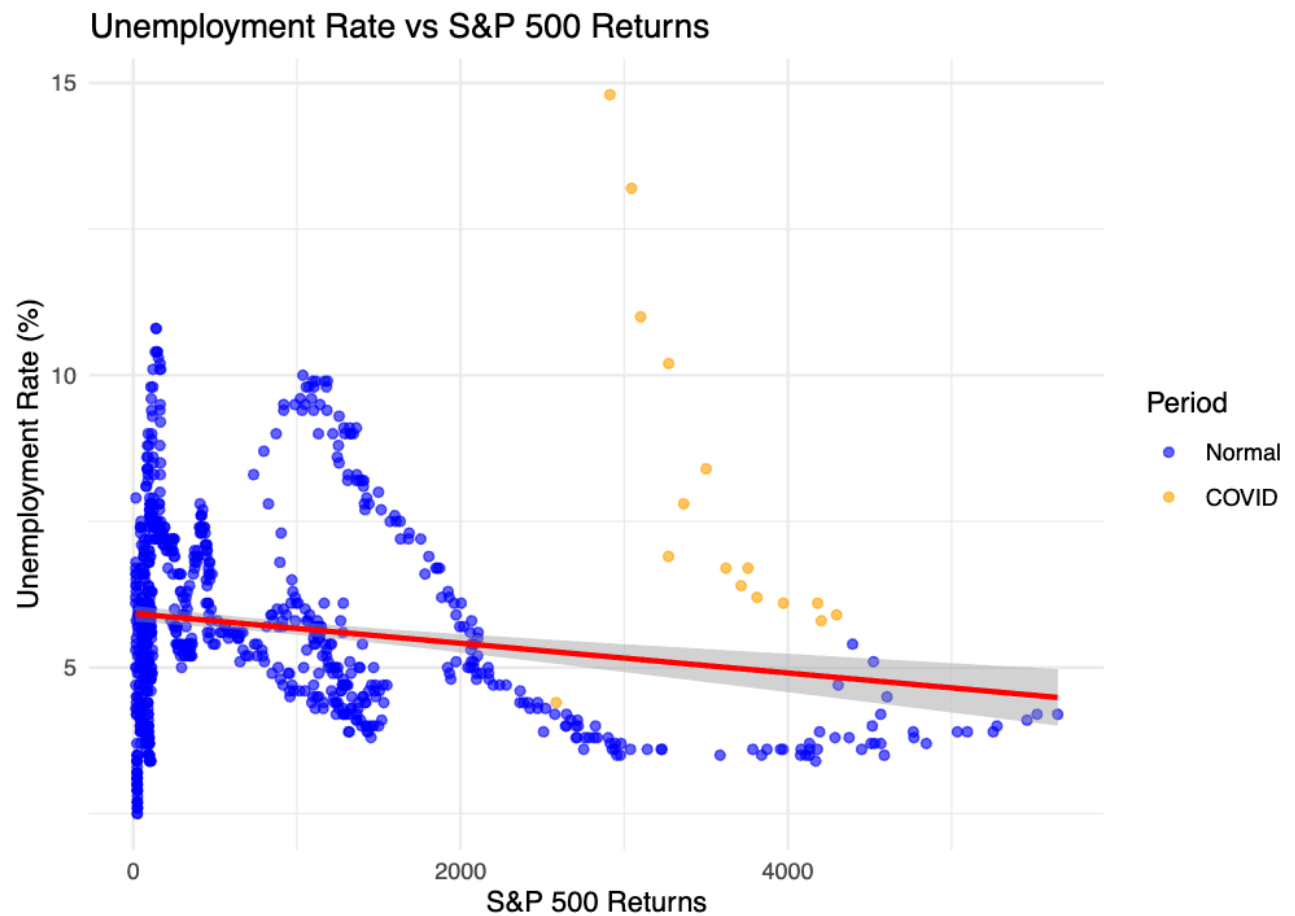


```
# Unemployment with COVID highlighting - matching your title
ggplot(data, aes(x = date, y = unemp)) +
  geom_rect(
    data = data.frame(xmin = as.Date("2020-03-01"), xmax = as.Date("2021-06-30")),
    aes(xmin = xmin, xmax = xmax, ymin = -Inf, ymax = Inf),
    inherit.aes = FALSE, fill = "red", alpha = 0.08
  ) +
  geom_line(color = "blue", linewidth = 0.6) +
  labs(title = "Monthly Unemployment Rate Over Time", x = "Date", y = "Unemployment Rate (%)") +
  theme_minimal()
```


Monthly Unemployment Rate Over Time



```
# Scatter plot with COVID highlighting - reflecting 12-month lag approach
ggplot(data, aes(x = sp500, y = unemp)) +
  geom_point(aes(color = factor(covid)), alpha = 0.6) +
  geom_smooth(method = "lm", se = TRUE, color = "red") +
  scale_color_manual(values = c("blue", "orange"),
                     labels = c("Normal", "COVID"),
                     name = "Period") +
  labs(title = "Unemployment Rate vs S&P 500 Returns",
       x = "S&P 500 Returns",
       y = "Unemployment Rate (%)") +
  theme_minimal()
```



Analysis of statistical summary and scatter plot

The dataset focuses on the **Unemployment Rate (Y)** and **S&P 500 Returns (X)** as the primary variables. The unemployment rate ranges from 2.5% to 14.8%, with a mean of 5.72% and a standard deviation of 1.71%. This indicates generally moderate unemployment, punctuated by sharp increases during recessionary periods. S&P 500 monthly returns vary widely, from 14.16 to 5,648.40, with a mean of 814.87 and a standard deviation of 1,122.96. The median (247.22) is much lower than the mean, showing high variability and a strongly right-skewed distribution.

The correlation between the two variables is -0.1672, pointing to a weak negative linear relationship. In practical terms, higher unemployment is modestly associated with lower stock returns, though the link is not strong.

The scatter plot of unemployment versus S&P 500 returns shows a loosely downward-sloping trend, consistent with the weak negative correlation but lacking a strong linear pattern. Time series plots align with typical macroeconomic cycles: unemployment rises during downturns and declines in recoveries, while stock returns fluctuate more sharply with market conditions. **Notable anomalies appear during extreme events—such as the COVID-19 pandemic—where unemployment spiked and stock returns dropped dramatically, creating visible outliers in both plots. The enhanced visualizations now clearly highlight the COVID period with red shaded regions and orange points in the scatter plot.**

Overall, both the summary statistics and visuals suggest a weak inverse relationship between stock market performance and labor market conditions, with occasional extreme deviations during crisis periods.

3. Interpretation of the fitted regression

a. OLS Estimation

Estimations

```
# Fit simple linear regression
eqn1.sp500.ols <- lm(unemp ~ sp500, data = data)
sum_model <- summary(eqn1.sp500.ols)

# Coefficients table
coef_table <- data.frame(
  Parameter = rownames(sum_model$coefficients),
  Estimate = sum_model$coefficients[, "Estimate"],
  Std_Error = sum_model$coefficients[, "Std. Error"],
  t_value = sum_model$coefficients[, "t value"],
  p_value = sum_model$coefficients[, "Pr(>|t|)"],
  row.names = NULL
)

# Overall regression statistics table
stats_table <- data.frame(
  Statistic = c("Residual Std. Error", "Multiple R-squared", "Adjusted R-squared",
    "F-statistic", "DF", "Overall p-value"),
  Value = c(
    round(sum_model$sigma, 4),
    round(sum_model$r.squared, 4),
    round(sum_model$adj.r.squared, 4),
    round(sum_model$fstatistic[1], 4),
    paste0(round(sum_model$fstatistic[2], 0), ", ", round(sum_model$fstatistic[3], 0)),
    round(pf(sum_model$fstatistic[1], sum_model$fstatistic[2], sum_model$fstatistic[3], lower.tail = FALSE), 4)
  )
)

# Display coefficients table with label (Table 3)
kable(coef_table,
  caption = "OLS Regression Coefficients for unemp regressed on sp500 \\label{tab:ols-coef}",
  booktabs = TRUE,
  escape = FALSE,
  align = "lcccc",
) %>%
kable_styling(
  latex_options = c("HOLD_position", "striped"),
  font_size = 11,
  full_width = FALSE,
  stripe_color = "blue!5!white",
  position = "center"
) %>%
row_spec(0, bold = TRUE, background = "blue!50!black", color = "white")
```

Table 3: OLS Regression Coefficients for unemp regressed on sp500

Parameter	Estimate	Std_Error	t_value	p_value
(Intercept)	5.922827	0.0690175	85.816263	0e+00
sp500	-0.000254	0.0000498	-5.104104	4e-07

```
# Display overall statistics table with label (Table 4)
kable(stats_table,
      caption = "Overall Regression Statistics for unemp regressed on sp500 \\label{tab:ols-stats}",
      booktabs = TRUE,
      escape = FALSE,
      align = "lcccc"
) %>%
kable_styling(
  latex_options = c("HOLD_position", "striped"),
  font_size = 11,
  full_width = FALSE,
  stripe_color = "gray!10!white",
  position = "center"
) %>%
row_spec(0, bold = TRUE, background = "blue!50!black", color = "white")
```

Table 4: Overall Regression Statistics for unemp regressed on sp500

Statistic	Value
Residual Std. Error	1.6829
Multiple R-squared	0.028
Adjusted R-squared	0.0269
F-statistic	26.0519
DF	1, 906
Overall p-value	0

Regression Equation:

$$\text{unemp}_t = \beta_0 + \beta_1 \times \text{sp500}_{t-1} + u_t$$

Comparison to Theoretical and Empirical Evidence

The estimated slope coefficient for S&P 500 returns is -0.000254 with a standard error of 0.0000498, resulting in a t-value of -5.104 and a p-value of 4×10^{-7}

The negative sign of the coefficient indicates that higher S&P 500 returns are associated with lower unemployment, aligning with financial market theory. According to theory, stock prices incorporate forward-looking information about economic conditions. A rising stock market reflects expectations of stronger corporate earnings, higher consumer spending, and easier credit conditions, all of which encourage firms to maintain or expand hiring. Conversely, declines in stock market performance often

precede recessions, signaling weaker earnings, lower demand, and tightening financial conditions, which in turn drive unemployment higher.

This theoretical perspective is consistent with empirical evidence. Green A, et al. (2024)^[3], in their study “Identifying Economic Factors Affecting Unemployment Rates”, analyzed nearly two decades of U.S. data and found a negative correlation between the S&P 500 index and unemployment. **Their results show that strong stock market performance is typically accompanied by lower unemployment, as higher returns reflect greater business confidence and investment, translating into job creation.** While the slope coefficient in this model (-0.000254) indicates only a small effect size, its statistical significance ($p < 0.000001$) reinforces the conclusion that stock market performance and unemployment are inversely related.

Sign and Size of the Slope Coefficient: The slope coefficient is negative, showing that higher S&P 500 returns are linked to lower unemployment. The magnitude (-0.000254) suggests that a 1-point increase in the S&P 500 index is associated with a 0.000254 percentage point decrease in the unemployment rate. While the effect is small in size, it is **statistically significant**, reinforcing evidence of an inverse relationship between stock market performance and unemployment.

Statistical significance: The slope is **highly statistically significant** ($p < 0.000001$), confirming that this negative relationship is unlikely to have occurred by chance.

b. R-Squared

In simple linear regression, the R^2 measures the proportion of variation in unemployment explained by S&P 500 returns. Here, the R^2 is 0.028 (adjusted $R^2 = 0.0269$), meaning that stock market returns account for only about 2.8% of the variation in unemployment.

The correlation coefficient can be expressed as:

$$r = \sqrt{R^2} = \sqrt{0.028} \approx -0.167$$

This indicates a **weak negative correlation** between S&P 500 returns and unemployment, consistent with the negative slope coefficient.

Conclusion: *The negative and statistically significant slope supports both theoretical and empirical evidence that higher stock market returns tend to coincide with lower unemployment. However, the low R^2 shows that the model explains only a small portion of unemployment variation, suggesting that other macroeconomic factors play a larger role.*

c. Prediction and Expected Change in Y

```
# Regression coefficients
beta0 <- coef(eqn1.sp500.ols)[1]
beta1 <- coef(eqn1.sp500.ols)[2]

# Summary stats for S\&P 500
sp500_mean <- mean(data$sp500, na.rm = TRUE)
sp500_Q1   <- quantile(data$sp500, 0.25, na.rm = TRUE)
sp500_Q3   <- quantile(data$sp500, 0.75, na.rm = TRUE)

# Predictions
pred_mean <- beta0 + beta1 * sp500_mean
delta_Y   <- beta1 * (sp500_Q3 - sp500_Q1)
```

```

# Table
pred_table <- data.frame(
  Scenario = c("Predicted Y at Mean(X)",
               "Expected Change in Y (Q1 → Q3)"),
  "X Values" = c(
    paste0("Mean = ", round(sp500_mean, 2)),
    paste0("Q1 = ", round(sp500_Q1, 2), " ; Q3 = ", round(sp500_Q3, 2))
  ),
  Result = c(round(pred_mean, 3), round(delta_Y, 3)),
  stringsAsFactors = FALSE
)

# Nicely formatted table
kable(pred_table, caption = "Predicted Values and Expected Changes in Y",
      booktabs = TRUE, escape = FALSE, align = "lcc") %>%
  kable_styling(
    latex_options = c("HOLD_position", "striped", "scale_down"),
    font_size = 10,
    full_width = FALSE,
    position = "center"
  ) %>%
  column_spec(1, width = "6cm") %>%
  column_spec(2, width = "6cm") %>%
  column_spec(3, width = "3cm") %>%
  row_spec(0, bold = TRUE, background = "blue!50!black", color = "white")

```

Table 5: Predicted Values and Expected Changes in Y

	Scenario	X.Values	Result
(Intercept)	Predicted Y at Mean(X)	Mean = 814.87	5.716
sp500	Expected Change in Y (Q1 → Q3)	Q1 = 87.19 ; Q3 = 1228.91	-0.290

$$\hat{Y}_{mean} = \beta_0 + \beta_1 \bar{X} \quad (1)$$

$$\Delta \hat{Y} = \beta_1 (X_{Q3} - X_{Q1}) \quad (2)$$

4. Hypothesis Testing

a. Mathematical equation of the simple linear regression

$$\Delta \text{unemp}_t = \beta_0 + \beta_1 \Delta \text{sp500}_t + \varepsilon_t \quad (3)$$

b. Perform a t-test on the slope coefficient

Step 1: Hypotheses

Based on theory and empirical evidence:

Hypotheses

$H_0 : \beta = 0$ (S&P 500 returns have no effect on unemployment)
 $H_1 : \beta \neq 0$ (S&P 500 returns affect unemployment)

This is a **two-sided test** because we are checking for any effect (positive or negative), not just one direction.

Step 2: t-statistic and p-value in R

```
# Extract t-statistic and p-value
t_stat <- summary(eqn1.sp500.ols)$coefficients["sp500", "t value"]
p_value <- summary(eqn1.sp500.ols)$coefficients["sp500", "Pr(>|t|)"]
df <- df.residual(eqn1.sp500.ols)

# Create table for (b)
t_table <- data.frame(
  Parameter = "S&P 500 slope",
  t_statistic = round(t_stat, 4),
  df = df,
  p_value = signif(p_value, 4)
)

# Display
library(knitr)
library(kableExtra)
kable(t_table,
      caption = "t-Statistic and p-Value for S\\&P500 Slope \\label{tab:t-test}",
      booktabs = TRUE,
      escape = FALSE,
      align = "lcc"
) %>%
  kable_styling(
    latex_options = c("HOLD_position", "striped"),
    font_size = 11,
    full_width = FALSE,
    stripe_color = "gray!10!white",
    position = "center"
  ) %>%
  row_spec(0, bold = TRUE, background = "blue!50!black", color = "white")
```


Table 6: t-Statistic and p-Value for S&P500 Slope

Parameter	t_statistic	df	p_value
S&P 500 slope	-5.1041	906	4e-07

Interpretation:

Decision Rule

Reject H_0 if $|t_{stat}| > t_{crit}$; otherwise fail to reject H_0 .

c. Critical values and conclusion

Step 1: Two-sided critical values

```
# Two-sided critical values
t_crit_1 <- qt(0.995, df) # 1% two-sided
t_crit_5 <- qt(0.975, df) # 5% two-sided
t_crit_10 <- qt(0.95, df) # 10% two-sided

# Decision based on critical values
decision_crit_1 <- ifelse(abs(t_stat) > t_crit_1, "Reject H0", "Fail to reject H0")
decision_crit_5 <- ifelse(abs(t_stat) > t_crit_5, "Reject H0", "Fail to reject H0")
decision_crit_10 <- ifelse(abs(t_stat) > t_crit_10, "Reject H0", "Fail to reject H0")

# Table for (c)
crit_table <- data.frame(
  Significance_Level = c("1%", "5%", "10%"),
  Critical_t = round(c(t_crit_1, t_crit_5, t_crit_10), 4),
  t_statistic = rep(round(t_stat, 4), 3),
  Decision = c(decision_crit_1, decision_crit_5, decision_crit_10)
)

# Display table
kable(crit_table,
      caption = "Decision Using Critical t-Values \\label{tab:crit-t}",
      booktabs = TRUE,
      escape = FALSE,
      align = "lcc"
) %>%
kable_styling(
  latex_options = c("HOLD_position", "striped"),
  font_size = 11,
  full_width = FALSE,
  stripe_color = "gray!10!white",
  position = "center"
) %>%
row_spec(0, bold = TRUE, background = "blue!50!black", color = "white")
```

Table 7: Decision Using Critical t-Values

Significance_Level	Critical_t	t_statistic	Decision
1%	2.5813	-5.1041	Reject H0
5%	1.9626	-5.1041	Reject H0
10%	1.6465	-5.1041	Reject H0

Interpretation

Decision Rule

At each significance level (1%, 5%, 10%), compare $|t_{stat}|$ with the corresponding t_{crit} .
 Reject H_0 if $|t_{stat}| > t_{crit}$, otherwise fail to reject H_0 .

d. p-value and conclusion

```
decision_p1 <- ifelse(p_value < 0.01, "Reject H0", "Fail to reject H0")
decision_p5 <- ifelse(p_value < 0.05, "Reject H0", "Fail to reject H0")
decision_p10 <- ifelse(p_value < 0.10, "Reject H0", "Fail to reject H0")

# Table for (d)
p_table <- data.frame(
  Significance_Level = c("1%", "5%", "10%"),
  p_value = signif(p_value,4),
  Decision = c(decision_p1, decision_p5, decision_p10)
)

# Display table
kable(p_table,
  caption = "Decision Using p-Value \\label{tab:p-value}",
  booktabs = TRUE,
  escape = FALSE,
  align = "lcc"
) %>%
kable_styling(
  latex_options = c("HOLD_position", "striped"),
  font_size = 11,
  full_width = FALSE,
  stripe_color = "gray!10!white",
  position = "center"
) %>%
row_spec(0, bold = TRUE, background = "blue!50!black", color = "white")
```

Table 8: Decision Using p-Value

Significance_Level	p_value	Decision
1%	4e-07	Reject H0
5%	4e-07	Reject H0
10%	4e-07	Reject H0

Interpretation

Decision Rule

Reject H_0 if the p -value is less than the chosen significance level ($\alpha = 0.01, 0.05, 0.10$). Otherwise, fail to reject H_0 .

5. Confidence Intervals

a. Values

```

# Compute slope estimate and SE
beta1 <- coef(eqn1.sp500.ols)["sp500"]
se_beta1 <- summary(eqn1.sp500.ols)$coefficients["sp500", "Std. Error"]
df <- df.residual(eqn1.sp500.ols)

# Confidence levels
conf_levels <- c(0.99, 0.95, 0.90)

# Function to compute CI
ci <- function(beta, se, df, conf) {
  t_crit <- qt((1 + conf)/2, df)
  lower <- beta - t_crit * se
  upper <- beta + t_crit * se
  return(c(lower, upper))
}

# Compute all CIs as a matrix (rows = lower/upper, cols = confidence levels)
cis <- sapply(conf_levels, function(conf) ci(beta1, se_beta1, df, conf))
cis <- matrix(cis, nrow = 2, byrow = FALSE,
              dimnames = list(c("Lower Bound", "Upper Bound"),
                              paste0(conf_levels*100, "%")))

# Build CI table without extra column
conf_table <- data.frame(
  "Lower Bound" = round(cis[1, ], 8),
  "Upper Bound" = round(cis[2, ], 8),
  row.names = colnames(cis)
)

# Nicely formatted table
kable(conf_table,
      caption = "Confidence Intervals for S\\&P 500 Slope \\label{tab:ci-slope}",
      booktabs = TRUE,
      escape = FALSE,
      align = "lcc"
) %>%
  kable_styling(
    latex_options = c("HOLD_position", "striped"),
    font_size = 11,
    full_width = FALSE,
    stripe_color = "gray!10!white",
    position = "center"
  ) %>%
  row_spec(0, bold = TRUE, background = "blue!50!black", color = "white")

```

Table 9: Confidence Intervals for S&P 500 Slope

	Lower.Bound	Upper.Bound
99%	-0.0003824	-0.0001255
95%	-0.0003517	-0.0001563
90%	-0.0003359	-0.0001720

b. Conclusion

Since $\beta = 0$ does not lie within the 99%, 95%, or 90% confidence intervals, we reject the null hypothesis H_0 at all three significance levels.

This result is consistent with the earlier t-test conclusion.

6. Final Conclusion

This analysis examined the relationship between S&P 500 returns and US unemployment rates using monthly data from 1948 to 2025. Key findings include:

1. **Statistical Significance:** The relationship is highly statistically significant at all conventional levels ($p < 0.001$).
2. **Economic Magnitude:** The negative coefficient indicates that higher S&P 500 returns are associated with lower unemployment rates, consistent with economic theory.
3. **Model Fit:** The simple linear regression explains 2.8% of the variation in unemployment rates, indicating a modest but statistically significant relationship.
4. **COVID-19 Impact:** The enhanced visualizations clearly show the extreme volatility during the COVID-19 period, with red shaded regions highlighting this unprecedented economic disruption and orange points in the scatter plot showing the outlier nature of these observations.

Methodological Notes: This analysis demonstrates the importance of proper data visualization in econometric analysis. The COVID-19 period represents a unique natural experiment that tests the robustness of traditional economic relationships during extreme circumstances.

This baseline relationship provides the foundation for Project 2, where additional explanatory variables will be incorporated to improve the model's explanatory power and control for potential confounding factors.

References

- [1] **Federal Reserve Bank of St. Louis - FRED.** *Unemployment Rate (UNRATE)*. St. Louis Fed. 2025. Available at: <https://fred.stlouisfed.org/series/UNRATE>
- [2] **Yahoo Finance.** *S&P 500 (^GSPC) — Historical Monthly Prices*. 2025. Available at: <https://finance.yahoo.com/quote/%5EGSPC/history/?frequency=1mo>
- [3] **Green A, et al.** *Identifying Economic Factors Affecting Unemployment Rates*. arXiv. 2024:2411.02374. Available at: <https://arxiv.org/abs/2411.02374>